

EIF2AK3/PERK Nuclear Localization

(GO:0005634) — Focused Curation Report

Gene: EIF2AK3 (PERK) · Human · UniProt Q9NZJ5 **Hypothesis (seed):** Human EIF2AK3/PERK localizes to the nucleus (GO:0005634). **Focus type:** function_assignment (does the gene product directly have GO:0005634?) **Date:** 2026-09-21 · Iteration 1

Executive Judgment

Verdict: Over-annotated / weakly supported (non-core at best).

The "nucleus" (GO:0005634) assignment for PERK is **not supported by any direct primary experimental localization evidence**. In UniProt/QuickGO (Q9NZJ5) the single GO:0005634 term carries the evidence code **IBA / GO_REF:0000033 (GO_Central/PANTHER PTN000113601)** — a *phylogenetically inferred* prediction, not an experimental result. Its `withFrom` set is seeded by soluble family paralogs that genuinely enter the nucleus — notably **PKR/EIF2AK2 (P19525)**, which carries **EXP** nucleus annotations

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and IDA
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and **HRI/EIF2AK1 (Q9BQI3)** with IDA. PERK itself (and GCN2/Q9P2K8) have **IBA-only** nucleus calls. Every experimentally- or manually-supported cellular-component annotation for PERK is ER-centric, and UniProt's manually curated *Subcellular location* comment is exclusively **"Endoplasmic reticulum membrane; Single-pass type I membrane protein."**

The one quasi-experimental nuclear signal for PERK itself is **Human Protein Atlas** immunofluorescence (Nucleoplasm/Cytosol main; Mitochondria additional). However this is single-source, only **"Approved"** reliability, and — tellingly — **HPA fails to detect the ER**, PERK's biochemically established compartment, which points to antibody-specificity limitations rather than a robust nuclear pool.

Two independent lines of reasoning argue the term is a **paralog/family over-annotation**:

1. **Topology.** PERK is an integral single-pass type-I ER-membrane protein (luminal sensor 30–514, TM 515–535, cytoplasmic kinase domain 536–1116). A membrane-anchored protein cannot reside free in the nucleoplasm. Any nuclear-adjacent pool would be the **nuclear envelope** (outer nuclear membrane, continuous with ER → GO:0005635), with the kinase domain facing the cytoplasm — **not** the nucleoplasmic interior that GO:0005634 denotes.
2. **Family inference.** The soluble eIF2 α -kinase paralogs do have genuine nuclear pools (PKR/EIF2AK2 in nucleus/nucleoli; GCN2). The GO_Central family tree therefore propagates a "nucleus" ancestral state that fits the soluble kinases but is inappropriate for the uniquely membrane-bound PERK.

Caveat / what would change the call: ER is continuous with the nuclear envelope, and PERK functionally interacts with nuclear-envelope-associated factors (e.g., PARP16). If a curator wishes to capture that, the correct term is **nuclear envelope (GO:0005635)** or ER, *not* nucleus (GO:0005634). I found no primary paper demonstrating nucleoplasmic PERK.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
UniProt Q9NZJ5 / QuickGO (database record; retrieved 2026-09-21)	Database/ annotation	Refutes (as direct evidence)	Is GO:0005634 experimentally supported for PERK?	GO:0005634 = IBA / GO_REF:0000033 (GO_Central) only; withFrom = PKR(P19525), HRI(Q9BQI3), GCN2 orthologs; curated CC comment = ER membrane, single-pass type I	Human, Swiss-Prot
QuickGO family check (P19525, Q9BQI3, Q9P2K8, Q9NZJ5)	Structural/ evolutionary	Competing / explains IBA	Which family members drive the nucleus IBA?	PKR has EXP nucleus (P 21029237) P 21072047 + IDA (P 26705305); HRI IDA; PERK & GCN2 IBA-only	Human paralogs
Human Protein Atlas (ENSG00000172071; IF)	Localization	Qualifies / weak-support	Does PERK show a nuclear pool?	Nucleoplasm+Cytosol (main); reliability "Approved"; no ER detected	Human cell lines, antibody IF
UniProt Q9NZJ5 topology features	Structural/ topology	Qualifies/ Refutes	Can full-length PERK be nucleoplasmic?	Signal 1–29; luminal 30–514; TM 515–535; cytoplasmic kinase 536–1116 → membrane-anchored	Human
	Localization (paralog)		Do family paralogs	"a fraction of PKR maps to the nucleoli"	

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 41406153 (2025)		Competing / explains IBA	localize to nucleus?		Human PKR/ EIF2AK2
P 29352251 (2018)	Localization (paralog)	Competing / explains IBA	PKR nuclear/ cytoplasmic partition	"repartition of PKR in the cytoplasm and the nucleus"	PKR/ EIF2AK2
P 34094832 (2021)	Interaction/ localization	Qualifies	PERK near nuclear envelope?	PARP16 "correlated with the nuclear envelope and the ER"; interacts with PERK/IRE1	Vascular cells
P 16352659 (2006)	Mutant phenotype	Qualifies (ER role)	PERK's cellular compartment of action	PERK—/— causes "distention and fragmentation of the ER"; ER Ca ²⁺ signalling defects	Mouse secretory/ muscle cells
P 26268696 (2015)	Interaction/ localization	Qualifies (ER role)	PERK compartment	PERK is an "ER transmembrane protein"; binds ER-membrane TMEM33	Breast cancer cells
P 16432136 (2006, review)	Review	Qualifies	ER–NE continuity	ER "is contiguous with the nuclear envelope"	Review

GO Decision Table

GO term	Aspect	Current evidence for PERK	Recommendation (lead)
GO:0005634 nucleus	CC	IBA-only (GO_REF:0000033), paralog-seeded; HPA IF nucleoplasm at "Approved", misses ER	Remove / down-weight — non-core, not directly supported
GO:0005635 nuclear envelope	CC	Not currently annotated; ER is continuous with NE; PARP16 interaction (P 34094832)	Optional add only if a specificity-validated primary source appears
GO:0005789 ER membrane	CC	ISS + NAS (P 11907036) + TAS(Reactome)	Retain (core)
GO:0005783 endoplasmic reticulum	CC	IDA (P 9930704), IC (P 11907036), TAS	Retain (core)
GO:0044233 MAM contact site	CC	IDA (P 39116259)	Retain
GO:0005829 cytosol	CC	TAS(Reactome) — cytosolic kinase- domain face	Retain (supportive)

Isoform / fragment check (Iteration 3)

UniProt Q9NZJ5 annotates **a single isoform, one mature chain (30–1116), no released cytosolic fragment**, and a crude scan found **no classical monopartite NLS** (≥ 4 consecutive K/R). The seed hypothesis's "relevant isoforms/fragments" route to the nucleus is therefore **not supported by curated sequence features**.

GO Curation Implications (leads — require curator verification)

- **GO:0005634 (nucleus, CC):** Recommend **remove or down-weight** as a stand-alone localization claim. It is IBA-only and conflicts with PERK's curated ER-membrane topology; it most likely reflects phylogenetic carry-over from the soluble paralogs (PKR/GCN2). Do **not** promote it to an experimental CC term.
 - **If a nuclear-adjacent location must be captured:** the biologically defensible term is **GO:0005635 (nuclear envelope)** or retention of the existing ER terms — but only if a curator finds a primary localization paper; I did not.
 - **Retain (well-supported):** GO:0005789 (ER membrane), GO:0005783 (ER, IDA), GO:0044233 (MAM contact site, IDA), GO:0005829/0005737 (cytosol/cytoplasm — the kinase domain's cytosolic face). These represent the gene product's primary location.
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Mechanistic Scope

The immediate molecular function of PERK is an **ER-membrane-resident eIF2 α protein kinase**: its luminal domain senses ER unfolded-protein load (via BiP release / direct binding), it oligomerizes/autophosphorylates, and its cytosolic kinase domain phosphorylates eIF2 α (Ser51) to attenuate translation and induce ATF4. **Nuclear consequences of the pathway (ATF4/CHOP transcription, NRF2 activation) are executed by downstream effectors that translocate to the nucleus — not by PERK itself.** Attributing "nucleus" to PERK conflates the sensor with its downstream transcriptional output.

Conflicts and Alternatives

- **Paralog confusion (most likely explanation):** PKR/EIF2AK2 has documented nuclear/nucleolar pools; GO_Central's family-level inference plausibly propagated a "nucleus" state to PERK.
- **ER-NE continuity artifact:** Immunofluorescence perinuclear ER signal can be misread as "nuclear." GO:0048471 (perinuclear region) is already annotated (IEA) and captures this better.

- **Topology constraint:** For membrane-anchored PERK, nucleoplasmic residence is topologically incoherent; only the nuclear envelope (ER-continuous) is physically available.

Knowledge Gaps

1. **Is there any *specific* primary paper showing nucleoplasmic PERK?** Checked UniProt CC, QuickGO, and PubMed; none found. HPA IF does report Nucleoplasm but at "Approved" reliability and without detecting the ER, so its antibody specificity is uncertain. Matters because HPA is currently the only experimental-type signal that could upgrade GO:0005634. *Resolve:* validate with a knockout-verified antibody (IF + subcellular fractionation), or confirm HPA antibody specificity against PERK-null cells.
2. **Does a proteolytically released PERK cytosolic fragment ever enter the nucleus?** Not established here. *Resolve:* look for cleavage/processing reports and fragment localization.
3. **Exact GO_Central IBA ancestor supporting the nucleus call.** Confirming it derives from PKR/GCN2 would formally document the over-annotation. *Resolve:* inspect the PAINT/GO_Central family tree.

Discriminating Tests

- **Subcellular fractionation + Western** (nuclear vs. ER/microsomal vs. cytosol) for endogenous PERK.
- **Confocal co-localization** of PERK with ER (calnexin), nuclear envelope (lamin B/emerin), and nucleoplasm markers, ± ER stress (thapsigargin/tunicamycin).
- **Human Protein Atlas** subcellular immunofluorescence review for EIF2AK3.
- **GO_Central/PAINT tree inspection** to confirm the IBA ancestor is a soluble nuclear paralog.

Curation Leads (verify before acting)

- **Action change:** Mark GO:0005634 (nucleus) as **not supported by direct evidence (IBA-only)**; do not treat as core localization. Prefer removal or replacement with GO:0005635 (nuclear envelope) *only if* a primary source is found.
- **Candidate references to cite for the ER core location:**

P 26268696

(PERK an ER transmembrane protein),

P 16352659

(PERK ER function/phenotype).

- **Candidate references explaining the over-annotation:**

P 41406153

P 29352251

(nuclear paralog PKR).

- **Suggested question for curator:** "Is the GO:0005634 term meant to capture nuclear-envelope/ER continuity? If so, GO:0005635 is more accurate; if nucleoplasm, no primary evidence exists."
- **Suggested experiment:** endogenous PERK fractionation + confocal, ± ER stress.

Provenance (computed, live queries 2026-09-21)

- **UniProt REST (Q9NZJ5):** curated CC = "Endoplasmic reticulum membrane; Single-pass type I membrane protein"; topology signal 1–29, lumenal 30–514, TM 515–535, cytoplasmic 536–1116; single isoform; single chain 30–1116; no NLS by crude K/R scan.
- **QuickGO annotation API (Q9NZJ5):** GO:0005634 = IBA / GO_REF:0000033 (GO_Central) only; `withFrom` includes PKR (P19525), HRI (Q9BQI3), GCN2 orthologs (PANTHER PTN000113601). Family check: PKR has **EXP** nucleus

P 21029237

P 21072047

+ IDA

(P 26705305);

HRI IDA; PERK & GCN2 IBA-only.

- **Human Protein Atlas API (ENSG00000172071):** Subcellular = Nucleoplasm/Mitochondria/Cytosol; main = Nucleoplasm/Cytosol; Reliability(IF) = "Approved"; **ER not reported** (specificity concern).
- **PubMed:** no primary evidence for nucleoplasmic PERK; nuclear pools confirmed for paralog PKR.

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